Machine Learning Evaluation Metrics Cheat Sheet

A comprehensive guide for data scientists with optimal value ranges and performance interpretations

Classification Metrics

Metric	Formula	Range	Excellent	Good	Fair	Poor	Use Case & Notes
Accuracy	(TP+TN)/(TP+TN+FP+FN)	[0,1]	≥0.90	0.80-0.89	0.70-0.79	<0.70	Balanced datasets only; misleading with imbalanced data
Precision	TP/(TP+FP)	[0,1]	≥0.90	0.80-0.89	0.70-0.79	<0.70	Minimize false positives (spam detection, fraud)
Recall (Sensitivity)	TP/(TP+FN)	[0,1]	≥0.90	0.80-0.89	0.70-0.79	<0.70	Minimize false negatives (medical diagnosis)
Specificity	TN/(TN+FP)	[0,1]	≥0.90	0.80-0.89	0.70-0.79	<0.70	True negative rate; important for screening tests
F1 Score	2×(Precision×Recall)/(Precision+Recall)	[0,1]	≥0.90	0.80-0.89	0.70-0.79	<0.70	Harmonic mean; balances precision & recall
AUC-ROC	Area under ROC curve	[0,1]	≥0.90	0.80-0.89	0.70-0.79	<0.70	0.5 = random; best for balanced datasets
AUC-PR	Area under Precision-Recall curve	[0,1]	≥0.90	0.80-0.89	0.70-0.79	<0.70	Better for imbalanced datasets than AUC-ROC

Metric	Formula	Range	Excellent	Good	Fair	Poor	Use Case & Notes
Log Loss	Cross-entropy loss	[0,∞)	≤0.10	0.10-0.30	0.30-0.50	>0.50	Lower is better; penalizes confident wrong predictions
MCC	Matthews Correlation Coefficient	[-1,1]	≥0.80	0.60-0.79	0.40-0.59	<0.40	Considers all confusion matrix elements
False Positive Rate	FP/(FP+TN)	[0,1]	≤0.10	0.10-0.20	0.20-0.30	>0.30	Should be minimized
False Negative Rate	FN/(FN+TP)	[0,1]	≤0.10	0.10-0.20	0.20-0.30	>0.30	Should be minimized

Regression Metrics

Metric	Formula/Description	Range	Excellent	Good	Fair	Poor	Characteristics & Notes
MAE	Mean Absolute Error: $(1/n)\sum\|y_i - \hat{y}_i\|$	[0,∞)	≈0	Small relative to range	Moderate	Large	Same units as target; robust to outliers
MSE	Mean Squared Error: $(1/n)\sum(y_i - \hat{y}_i)^2$	[0,∞)	≈0	Small relative to range	Moderate	Large	Penalizes large errors; units squared
RMSE	Root Mean Squared Error: $\sqrt{MSE}$	[0,∞)	≈0	Small relative to range	Moderate	Large	Same units as target; sensitive to outliers
R^2	R-squared: 1 - (SS_res/SS_tot)	(-∞,1]	≥0.90	0.70-0.89	0.50-0.69	<0.50	Proportion of variance explained
Adjusted R^2	R^2 adjusted for number of predictors	(-∞,1]	≥0.90	0.70-0.89	0.50-0.69	<0.50	Penalizes unnecessary predictors
MAPE	Mean Absolute Percentage Error	[0,∞)	<5%	5-15%	15-25%	>25%	Percentage error; undefined when actual=0
RMSLE	Root Mean Squared Log Error	[0,∞)	≈0	Small	Moderate	Large	Good for skewed targets; log scale

Metric	Formula/Description	Range	Excellent	Good	Fair	Poor	Characteristics & Notes
MASE	Mean Absolute Scaled Error	[0,∞)	<1	1-2	2-3	>3	Compares to naive seasonal forecast

Clustering Metrics

Metric	Description	Range	Excellent	Good	Fair	Poor	Interpretation & Usage
Silhouette Score	(b-a)/max(a,b) averaged across samples	[-1,1]	>0.70	0.50-0.70	0.25-0.50	<0.25	Measures cluster cohesion and separation
Calinski-Harabasz Index	Between/within cluster variance ratio	[0,∞)	>300	100-300	50-100	<50	Higher is better; computationally efficient
Davies-Bouldin Index	Average similarity between clusters	[0,∞)	<0.50	0.50-1.00	1.00-2.00	>2.00	Lower is better; measures cluster similarity
Adjusted Rand Index	Similarity corrected for chance	[-1,1]	>0.80	0.50-0.80	0.20-0.50	<0.20	Requires true labels; corrects for chance
Mutual Information	Information shared between labels	[0,∞)	>0.80	0.50-0.80	0.20-0.50	<0.20	Requires true labels; normalized versions available
Inertia (WCSS)	Sum of squared distances to centroids	[0,∞)	Low	Moderate	High	Very High	Lower is better; use with elbow method

Time Series & Forecasting Metrics

Metric	Formula/Description	Range	Excellent	Good	Fair	Poor	Best Use Case & Limitations
MAPE	Mean Absolute Percentage Error	[0,∞)	<5%	5-10%	10-20%	>20%	Stable, non-zero values; biased toward under-forecasts
WAPE	Weighted Absolute Percentage Error	[0,∞)	<5%	5-10%	10-20%	>20%	Intermittent demand; weights by volume
WMAPE	Weighted Mean Absolute Percentage Error	[0,∞)	<5%	5-10%	10-20%	>20%	Allows different importance weights
SMAPE	Symmetric Mean Absolute Percentage Error	[0,200]	<10%	10-20%	20-50%	>50%	Handles near-zero values better than MAPE

Metric	Formula/Description	Range	Excellent	Good	Fair	Poor	Best Use Case & Limitations
MFE	Mean Forecast Error	(-∞,∞)	≈0	Small bias	Moderate bias	Large bias	Measures forecast bias; should be near zero
Tracking Signal	Cumulative forecast error/MAD	(-∞,∞)	[-2,2]	[-4,4]	[-6,6]	Outside [-6,6]	Monitors forecast bias over time

Model Selection Guidelines & Best Practices

Problem Type Framework

- **Classification** → Accuracy, Precision, Recall, F1, AUC-ROC/PR
- **Regression** → MAE, RMSE, R^2, MAPE
- **Clustering** → Silhouette, Calinski-Harabasz, Davies-Bouldin
- **Time Series** → MAPE, WAPE, MFE, Tracking Signal

Data Characteristics Considerations

Data Type	Recommended Metrics	Avoid	Notes
Balanced Classification	Accuracy, AUC-ROC, F1	-	Standard metrics work well
Imbalanced Classification	Precision, Recall, F1, AUC-PR	Accuracy	Focus on minority class performance
Cost-Sensitive	Custom metrics aligned with costs	Generic metrics	Weight by business impact
Regression with Outliers	MAE, Robust metrics	MSE, RMSE	Avoid metrics sensitive to outliers
Skewed Regression Targets	RMSLE, Log-transformed metrics	Standard metrics	Use log scale for better performance
Time Series	MAPE, WAPE, directional accuracy	Static metrics	Consider temporal dependencies

Cross-Validation Strategies

Problem Type	CV Method	K Value	Special Considerations
General Classification/Regression	K-Fold	5 or 10	Stratify for imbalanced data
Imbalanced Data	Stratified K-Fold	5 or 10	Maintains class distribution
Time Series	Time Series Split	Variable	Forward chaining; no data leakage
Small Datasets	Leave-One-Out	n	Computationally expensive
Large Datasets	Hold-out	-	70-80% train, 20-30% test

Red Flags & Troubleshooting

Issue	Symptoms	Likely Causes	Solutions
Perfect Scores	All metrics = 1.0 or 0.0	Data leakage, overfitting	Check feature engineering, validation
Inconsistent Metrics	Conflicting results	Wrong metric choice	Use appropriate metrics for problem
No Baseline	No comparison point	Missing baseline	Compare to simple baseline (mean, mode)
Overfitting	Train >> Test performance	Model too complex	Regularization, more data, simpler model
Underfitting	Both train/test poor	Model too simple	More features, complex model, less regularization

Universal Principles

1. **Multiple Metrics**: Never rely on single metric evaluation
2. **Domain Alignment**: Choose metrics matching business objectives
3. **Baseline Comparison**: Always compare against simple baselines
4. **Statistical Significance**: Use proper tests for model comparison
5. **Context Dependency**: Optimal values vary by domain and use case

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